



Crystal structure of a putative 3-hydroxypimelyl-CoA dehydrogenase, Hcd1, from *Syntrophus aciditrophicus* strain SB at 1.78 Å resolution

David M. Dinh,^{a,b} Leonard M. Thomas^{b,c} and Elizabeth A. Karr^{a,b*}

^aDepartment of Microbiology and Plant Biology, University of Oklahoma, 770 Van Vleet Oval, Norman, OK 73019, USA, ^bPrice Family Foundation Institute of Structural Biology, University of Oklahoma, 101 Stephenson Parkway, Norman, OK 73019, USA, and ^cDepartment of Chemistry and Biochemistry, University of Oklahoma, 101 Stephenson Parkway, Norman, OK 73019, USA. *Correspondence e-mail: lizkarr@ou.edu

Received 28 March 2023

Accepted 20 May 2023

Edited by R. L. Stanfield, The Scripps Research Institute, USA

Keywords: SDR-family proteins; Hcd1; syntrophy; fatty-acid metabolism; *Syntrophus aciditrophicus*.

PDB reference: Hcd1, 7sub

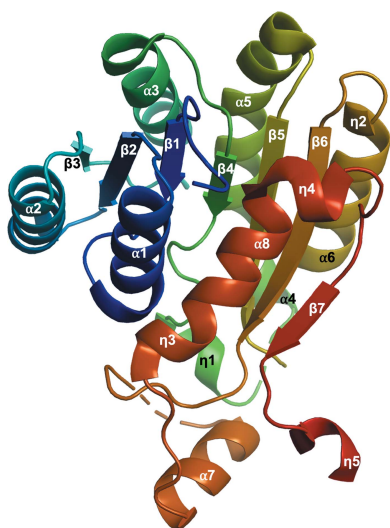
Supporting information: this article has supporting information at journals.iucr.org/f

Syntrophus aciditrophicus strain SB is a model syntroph that degrades benzoate and alicyclic acids. The structure of a putative 3-hydroxypimelyl-CoA dehydrogenase from *S. aciditrophicus* strain SB (*SaHcd1*) was resolved at 1.78 Å resolution. *SaHcd1* contains sequence motifs and structural features that belong to the short-chain dehydrogenase/reductase (SDR) family of NADPH-dependent oxidoreductases. *SaHcd1* is proposed to concomitantly reduce NAD⁺ or NADP⁺ to NADH or NADPH, respectively, while converting 3-hydroxypimelyl-CoA to 3-oxopimelyl-CoA. Further enzymatic studies are needed to confirm the function of *SaHcd1*.

1. Introduction

Aromatic compounds are one of the most common carbon sources for microorganisms, and anaerobic degradation of these compounds is an important part of the global carbon cycle (Evans, 1977). *Syntrophus aciditrophicus* strain SB is a model syntroph that degrades benzoate and cyclohexane-1-carboxylate to acetate and synthesizes benzoate and cyclohexane-1-carboxylate from crotonate (Moultaki *et al.*, 2007, 2008; Elshahed *et al.*, 2001; Elshahed & McInerney, 2001). Previous proteomic studies have shown that the degradation and formation of benzoate and cyclohexane-1-carboxylate in *S. aciditrophicus* share a core set of enzymes (James *et al.*, 2019). One step in this pathway is the conversion of 3-hydroxypimelyl-CoA to 3-oxopimelyl-CoA. *SaHcd1*, encoded by SYN_RS14115 (*S. aciditrophicus hcd1*), was annotated as a putative 3-hydroxypimelyl-CoA dehydrogenase and proposed to convert 3-hydroxypimelyl-CoA to 3-oxopimelyl-CoA (James *et al.*, 2019).

Short-chain dehydrogenase/reductase (SDR) superfamily proteins are a vast family spread across all three domains of life (Gabrielli *et al.*, 2023; Kallberg *et al.*, 2002, 2010). The SDR superfamily can be divided into numerous subfamilies (Persson *et al.*, 2009). Classical SDR-family proteins are, on average, 250 amino acids in length, and a second group is extended by an additional 100 amino-acid residues. Often composed of a single domain, the N-terminus houses the cofactor-binding motif and exhibits overall higher sequence similarity, whereas the C-terminus is more divergent and houses the substrate-binding region for the diverse biological reactions that this group of proteins carries out in nature (Kavanagh *et al.*, 2008). The hallmarks of SDR-family proteins include the dinucleotide-binding Rossmann fold, which houses a glycine-rich motif that, albeit with some variation,



conforms to a TGXXGXXG motif. Finally, the catalytic site of SDR-family proteins may utilize a catalytic tetrad motif (Asn–Ser–Tyr–Lys) or a triad motif (Ser–Tyr–Lys) (Filling *et al.*, 2002; Gabrielli *et al.*, 2023; Kallberg *et al.*, 2002; Kavanagh *et al.*, 2008).

SaHcd1 is a classical SDR protein containing the TGXXGXXG sequence motif and the Ser–Tyr–Lys catalytic triad typical of the superfamily and belongs to the cP1 subfamily (Persson *et al.*, 1991; Kallberg *et al.*, 2002). This study reports the structure of *SaHcd1* determined at 1.78 Å resolution and highlights a few distinct differences from closely related SDR-family proteins. Successful crystallization of *SaHcd1* required nonconventional protein-handling methods during purification. Structural information on *SaHcd1* provides greater insight into SDR-family proteins and represents the first structure of an SDR-family protein from a syntrophic bacterium.

2. Materials and methods

2.1. Protein cloning and expression

Full-length putative 3-hydroxypimelyl-CoA dehydrogenase (NCBI Reference Sequence WP_011418898.1) was PCR-amplified from genomic DNA using the primers shown in Table 1. DNA amplification was performed using Phusion High-Fidelity DNA Polymerase (New England Biolabs) following the manufacturer's directions. The resulting amplicon was cloned into expression vector pQE80LN, a pQE80L vector (Qiagen) modified to allow in-frame cloning of an N-terminal hexahistidine tag without an RGS epitope (Table 1). The amplicon and vector were digested with BamHI and PstI (New England Biolabs). The initial transformation was into DH5α to confirm inserts of appropriate size and sequence. After sequence confirmation by Sanger sequencing, the resulting plasmid was transformed into *Escherichia coli* Rosetta (BL21 derivative, DE3) cells. The cells were grown to an OD₆₀₀ of 0.5 at 37°C and 250 rev min^{−1}. To overexpress *SaHcd1*, the temperature was reduced to 18°C and isopropyl β-D-1-thiogalactopyranoside was added to a final concentration of 0.1 mM. The cells were shaken at 250 rev min^{−1} for an additional 18 h. The cells were pelleted at 14 000 rev min^{−1} for 15 min and the cell pellet was frozen at −80°C. The cells were resuspended in 50 mM Tris, 100 mM NaCl pH 8 and lysed using sonication, and the cell-free extract was frozen. Freezing of the cell lysate before purification was essential for successful crystallization. The amino-acid sequence of the expressed construct is provided in Table 1.

2.2. Protein purification

Nickel-affinity chromatography was carried out using a 25 ml gravity-flow column (Bio-Rad) and nickel nitrilotriacetic acid (Ni–NTA) agarose beads (MCLAB) at room temperature. 1 ml of beads was washed with distilled deionized water and subsequently with equilibration buffer (50 mM Tris, 100 mM NaCl pH 8). The previously frozen cell-free extract was added to the column and the beads were

Table 1

Protein-preparation information.

Source organism	<i>Syntrophus aciditrophicus</i> strain SB
DNA source	Genomic DNA provided by Dr Michael McInerney, University of Oklahoma, USA
Forward primer	5′-GATATTGGATCCATGGGAAAGTTAGATGGTAAG GTA-3′
Reverse primer	5′-TAGTAGCTGCAGGGTCAAACCATAACCACC-3′
Expression vector	A modified Qiagen pQE80L vector that changes the N-terminal added sequence from MRGSHHHHHHGS to MHHHHHHHGS
Expression host	<i>E. coli</i> Rosetta (BL21 derivative, DE3)
Expression construct amino-acid sequence	MHHHHHHHGS MGKLDGKVA VVTGGRGV GRAICVAY AEEGADV VVNYAGNHAAANEV VAMIEKMGRRAVA VQGRVEIKAEAEKTIQTAVDNFGRIDVLVNNAGA TKPAMLHKMTTEEQWDAVVNIHLKGFPLCVQAAAK YFMEQNYGKIINVT SVAGLVGTTGQINYSAAKGG ILSFTMSAARELARNVNTSNVISLGIVTDMTEK ITDDKLKEIYMRILLNRYAEPSEVAPAFVFFG CDDSKYITGQLKVDGGYGLTLQPSLIS

allowed to settle. The unbound material (flowthrough) was collected. During the initial purification, the column with the protein-bound beads was inadvertently frozen at −20°C and subsequently thawed. Ultimately, this inadvertent freezing step was found to be essential for successful crystallization. The column was then washed with 1 ml equilibration buffer followed by 1 ml 10, 20 and 50 mM imidazole in 50 mM Tris, 100 mM NaCl pH 8. Recombinant *SaHcd1* was eluted with 1 ml 60, 70, 80 and 90 mM imidazole in 50 mM Tris, 200 mM NaCl pH 8 and 1 ml 100, 150, 200, 300 and 500 mM imidazole in 50 mM Tris, 300 mM NaCl pH 8. All fractions were stored overnight at 4°C, which was another step that was found to be instrumental to successful crystallization. The eluates from 150, 200, 300 and 500 mM imidazole in 50 mM Tris, 300 mM NaCl pH 8 were combined and concentrated to approximately 600 µl using an Amicon Ultra-4 centrifugal unit (5 ml) with Ultracel-10 (10 kDa molecular-weight cutoff) membrane (EMD Millipore) following the manufacturer's guidelines. The samples were mixed intermittently by gentle pipetting inside the filter chamber.

The concentrated sample was purified by size-exclusion chromatography (SEC) at the University of Oklahoma Protein Production and Characterization Core Facility using a Superdex S200 Increase 10/300 GL column on an ÄKTApure M1 (GE Healthcare) with 20 mM Tris, 150 mM NaCl pH 8. SEC chromatograms (Supplementary Fig. S1a) were generated with UNICORN (GE Healthcare). Fractions were visualized by SDS–PAGE (Supplementary Fig. S1b) and fractions 13–15 (Supplementary Fig. S1a) containing recombinant *SaHcd1*, which elutes as a multimeric complex corresponding to a dimer, were pooled and concentrated to approximately 4–7 mg ml^{−1}. Protein concentrations were measured on a NanoDrop One C (Thermo Fisher Scientific). The Thermo Fisher Scientific PageRuler Plus pre-stained protein ladder was used for SDS–PAGE.

2.3. Protein crystallization

In attempts to initially crystallize *SaHcd1*, the commercial screens MCSG-1 and MCSG-2 (Antrace), the Berkeley Screen

Table 2

Summary of crystal parameters and data-collection, model and refinement statistics.

Values in parentheses are for the outermost resolution shell.

Resolution range (Å)	32.74–1.78 (1.84–1.78)
Space group	<i>P</i> 3 ₁ 21
<i>a</i> , <i>b</i> , <i>c</i> (Å)	56.01, 56.01, 133.13
α , β , γ (°)	90, 90, 120
Unique reflections	23968 (2349)
Completeness (%)	99.94 (99.96)
Multiplicity	9.7 (8.9)
Wilson <i>B</i> factor (Å ²)	30.40
<i>R</i> _{merge}	0.038 (0.84)
<i>R</i> _{meas}	0.04 (0.89)
<i>R</i> _{p.i.m.}	0.017 (0.40)
CC _{1/2}	0.99 (0.86)
Mean <i>I</i> / σ (<i>I</i>)	31.1 (2.5)
Reflections used in refinement	23963 (2348)
Reflections used for <i>R</i> _{free}	1995 (192)
<i>R</i> _{work}	0.18 (0.27)
<i>R</i> _{free}	0.21 (0.33)
No. of non-H atoms	
Total	1817
Macromolecules	1689
Ligands	6
Solvent	122
Protein residues	222
Root-mean-square deviations	
Bond lengths (Å)	0.02
Angles (°)	1.24
Ramachandran favored (%)	97.66
Ramachandran allowed (%)	2.34
Ramachandran outliers (%)	0.00
Rotamer outliers (%)	0.57
Clashscore	6.71
Average <i>B</i> factor (Å ²)	
Overall	37.70
Macromolecules	37.54
Ligands	40.24
Solvent	39.69

(Rigaku Reagents) and PEG/Ion HT (Hampton Research) were used. Condition and optimization variables included the protein concentration, well component concentrations, microseeding and the inclusion of NADH. Subsequently, *SaHcd1* crystallized in various conditions with little impact on the data quality. The crystals typically appeared as rhombohedrons (Supplementary Fig. S1c), with a small number appearing as triangular prisms. The final structure of *SaHcd1* was solved using crystals obtained at room temperature by mixing 1 µl reservoir solution consisting of 0.1 *M* sodium malonate pH 4.0, 12% (w/v) PEG 3350 with an equal volume of 4–7 mg ml^{−1} protein solution using the sitting-drop method. This condition is the same as condition 49 (well E1) from the PEG/Ion HT screen (Hampton Research). For cryoprotection, the crystals were soaked in buffer containing the mother liquor with 5–15% glycerol and were subsequently plunged into liquid nitrogen before storage.

2.4. Data collection and structure determination

Diffraction data for the *SaHcd1* crystal were collected at 100 K on experimental station 9-2 at the Stanford Synchrotron Radiation Lightsource, Menlo Park, California, USA using a Dectris PILATUS 6M pixel-array detector. The data set was processed using the *XDS* package (Kabsch, 2010). The

SaHcd1 structure was determined by molecular replacement with an ensemble (superposition of multiple homologous structures) of four structures using *Ensembler* from *Phenix* (Liebschner *et al.*, 2019): 3-oxoacyl-(acyl carrier protein) reductase from *Bacillus anthracis* strain Ames (PDB entry 2uvd, 46.80% identity; Zaccai *et al.*, 2008), FabG from *Synechocystis* sp. PCC 6803 (PDB entry 4rzh, 46.77% identity; Liu *et al.*, 2015), OAR (FabG) from *Listeria monocytogenes* (PDB entry 4jro, 45.90% identity; Center for Structural Genomics of Infectious Diseases, unpublished work) and ORF 1438 from *Clostridium thermocellum* (PDB entry 2hq1, 42.45% identity; Southeast Collaboratory for Structural Genomics, unpublished work). The model was refined using *phenix.refine* (Liebschner *et al.*, 2019) and the residue positions were manually adjusted in *Coot* (Emsley & Cowtan, 2004; Emsley *et al.*, 2010). The data-collection and refinement statistics are summarized in Table 2. The models were visualized and analyzed using *PyMOL* (DeLano, 2002). The atomic coordinates have been deposited in the Protein Data Bank (PDB entry 7sub).

3. Results and discussion

Crystallization of *SaHcd1* requires freezing and thawing the cell-free extract and a subsequent freeze–thaw cycle after loading the sample onto the nickel-affinity column. After freezing, the frozen column containing the sample can be thawed at any time. It was also necessary to store the eluates from nickel-affinity chromatography overnight at 4°C before concentrating the sample for size-exclusion chromatography. Attempts to crystallize *SaHcd1* without these additional steps resulted in no crystal formation. Additional empirical testing was not performed to determine whether a second freeze–thaw cycle of the cell-free extract versus freezing on the column was sufficient for successful crystallization. The lysis and purification steps used a temperature-sensitive Tris-based buffer system (Sambrook, 2001). Thus, the protein was subjected to multiple rounds of shifting pH with decreasing temperatures. It has also been established that freezing leads to temporary localized increases in solute concentrations and potential phase separations in the protein environment. Multiple rounds of freezing or rapid decreases in temperature are also known to lead to reversible partial protein unfolding or destabilization through degradation (Bhatnagar *et al.*, 2007; Strambini & Gabellieri, 1996). However, only minimal density was missing; thus, degradation was not likely to be a large contributor to successful crystallization. It is plausible that repeated shifts in the localized protein environment or partial unfolding and refolding somehow contribute to the overall propensity of *SaHcd1* to crystallize. While the techniques leading to crystallization were not the primary focus of this study, it warranted discussion that successful crystallization was only achieved with multiple rounds of cooling and warming of the sample, as it may be an avenue for other researchers to pursue when attempting to crystallize a protein.

The apo structure of *SaHcd1* has been determined at a resolution of 1.78 Å (Table 2). The model has a working *R*

factor of 0.1830 with a free R factor of 0.214 (Table 2). The chain is composed of eight α -helices (three disordered loops may potentially contain an additional α -helix) flanking a central seven-stranded parallel β -sheet forming the NAD(P)-binding Rossmann-like domain typical of SDR proteins (Figs. 1 and 3a). The final model contains a single chain in the unit cell in space group $P3_121$. The chain was not completely traced, with three unresolved loops missing four, 15 and nine residues (Fig. 2, red bars). No electron density was observed for the first ten N-terminal amino acids of the expression construct and the seven C-terminal residues (shown in italics in Table 1).

The DALI server (Holm, 2020) reveals close structural similarity and amino-acid identity to many other SDR proteins. R.m.s.d. values ranged from 2.2 to 2.7 Å and amino-acid identity ranged from 39% to 45% for the SDR-family protein structures used for molecular replacement and overall investigation (Supplementary Table S1, Fig. 2). The highest similarity was to a 3-oxoacyl-(acyl carrier protein) reductase from *B. anthracis* (PDB entry 2uvd; Zaccai *et al.*, 2008) and the lowest within the comparison group was to FabG from *Synechocystis* sp. PCC 6803 (PDB entry 4rzh; Liu *et al.*, 2015). An amino-acid alignment between *SaHcd1* and the comparison structures is shown in Fig. 2, with notable features highlighted.

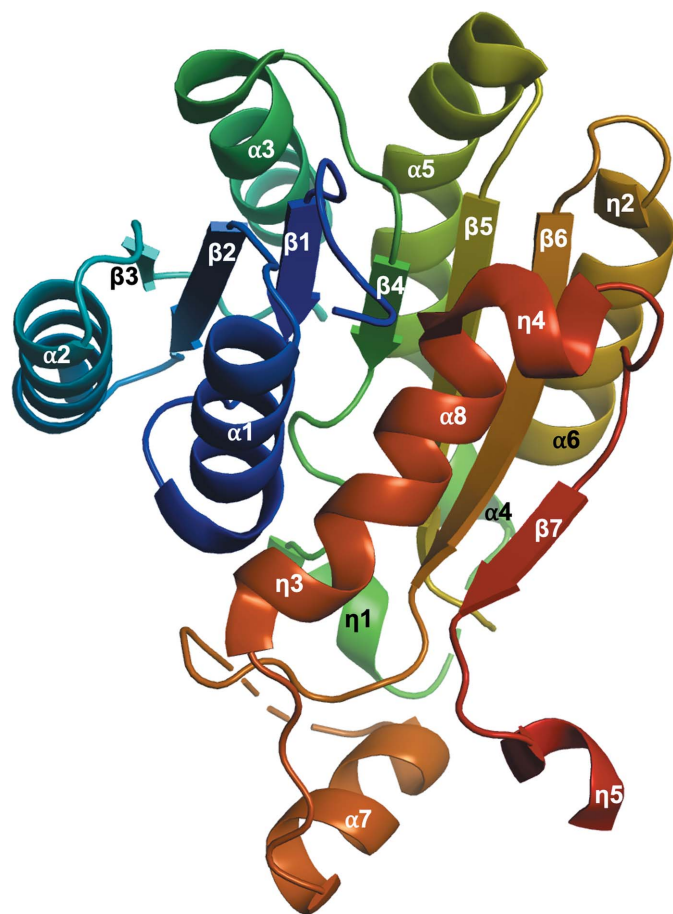


Figure 1
The overall structure of the *SaHcd1* monomer. The model is colored from dark blue at the N-terminus to red at the C-terminus, and secondary-structural elements (β -strands, 3_{10} - η -helices and α -helices) are labeled.

For viewing simplicity, PDB entries 2uvd and 4rzh will be used to visualize three distinct differences in the *SaHcd1* structure (Figs. 3b–3d) from other closely related structures listed in Supplementary Table S1.

The first distinct area of difference in the structure of *SaHcd1* occurs in the region of amino acids 90–120 encompassing the end of β_4 , 3_{10} - η -helix η_1 and α_4 (Figs. 2, 3b and 3c). An amino-acid alignment of this region shows that a well conserved aspartic acid and asparagine are replaced by a proline at position 97 and a histidine at position 115 in *SaHcd1*, respectively (Fig. 2, orange squares; Figs. 3b and 3c). The 3_{10} - η -helix η_1 seen in *SaHcd1* (Figs. 3a–3c) is not a feature of other closely related SDR protein structures, with this region being an unstructured loop between β_4 and α_4 (Figs. 3a–3c). The proline at residue 97 may partly be responsible for the altered structure in this region. Overall, there is a shortening of the helices that is not seen in other SDR-family proteins (Fig. 3b). Notably, density was missing for amino acids 104–107 (Fig. 2, red bar; Fig. 3b). Nevertheless, density was good for the surrounding residues, demonstrating an overall shifted position of the helices (Fig. 3b). The overall sequence similarity in this region is not highly conserved (Fig. 2, orange rectangle; Fig. 3a). Notably, the difference in this region correlates to a loss of the order of the N-terminal region of α_6 (Fig. 2, red bar). In the comparison structures, except for PDB entry 2hq1, the ordered helical region corresponding to *SaHcd1* α_6 is longer and is tightly packed between two helices C-terminal to β_5 (Fig. 3b, Supplementary Fig. S3). Just N-terminal to this region (amino acids 89–93) is a highly conserved region with the sequence NNAG that is present across all compared SDR proteins (Fig. 2, purple rectangle, Fig. 3a). Filling and coworkers suggest that these residues are critical to creating a structural ‘framework’ for positioning the cofactor (Filling *et al.*, 2002).

When superposed with the NAD from PDB entry 4nbt, it can be seen that the shifted region (Fig. 3b) would overlap with the cofactor-binding region (Fig. 3c, dashed circle). *E. coli* FabG is a rare example of an SDR-family protein that undergoes a conformational rearrangement of the cofactor-binding pocket upon binding NADP⁺. In the case of *E. coli* FabG, the region containing the serine of the Ser–Tyr–Lys catalytic triad moves to allow binding of NADP⁺ (Price *et al.*, 2001, 2004). Therefore, it remains plausible that *SaHcd1*, like FabG, undergoes a conformational change that would rearrange this region upon binding of the cofactor.

In *SaHcd1*, α_7 is shifted $\sim 90^\circ$ relative to the corresponding helices in the comparison structures (Fig. 3d). This region has little to no sequence conservation, but has overall structural conservation in the comparison structures except for PDB entry 4jro. In PDB entry 4jro the helix is extended by several residues. Not being involved in cofactor binding, the C-terminal region, except for the extreme C-terminus (Fig. 2), is noted for a lack of sequence conservation and involvement in binding the broad range of substrates known for SDR-family proteins (Gabielli *et al.*, 2023).

Based on sequence alignment (Fig. 2), *SaHcd1* contains the characteristic TGXXGXG glycine-rich motif. The positions

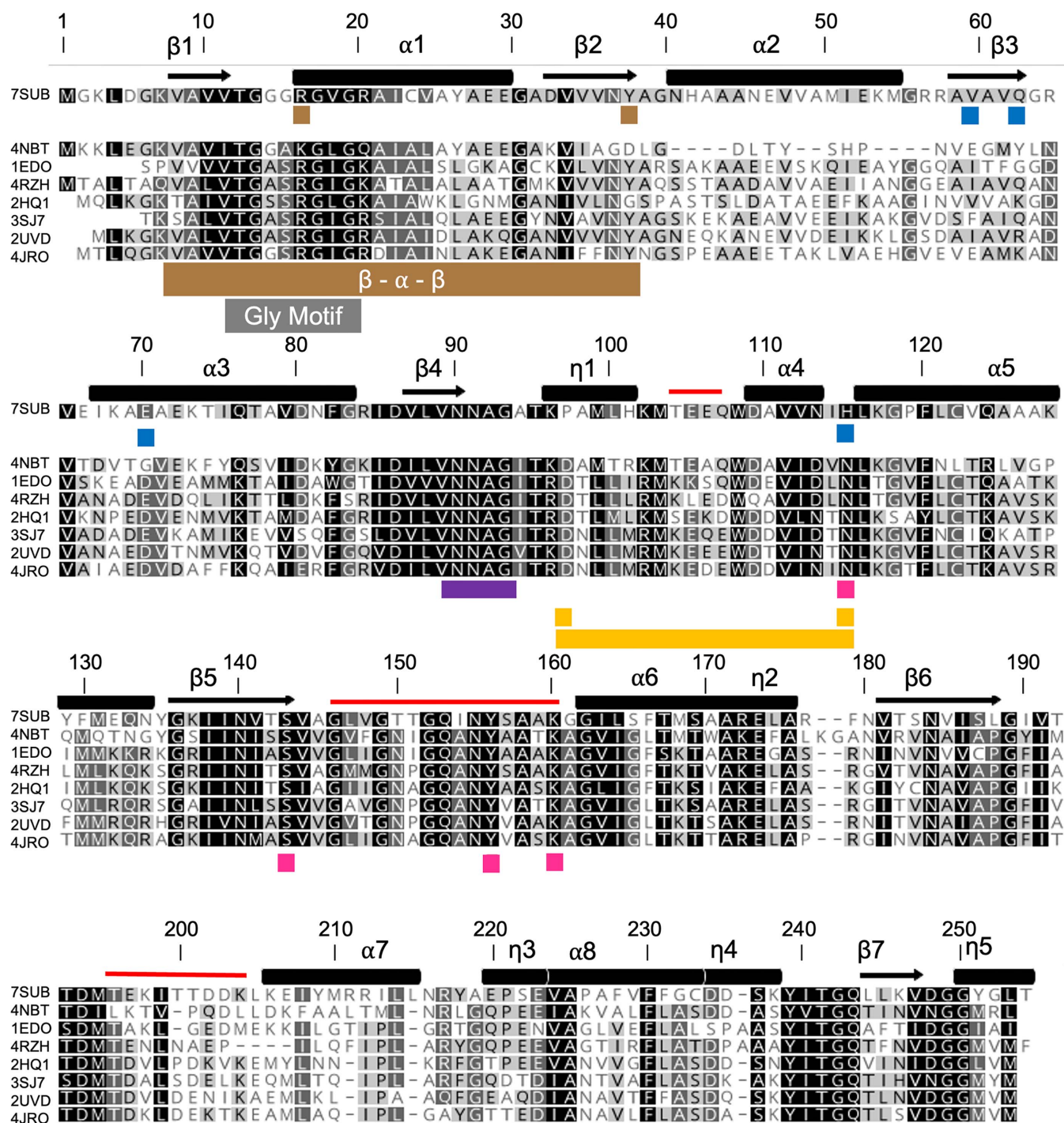


Figure 2

Amino-acid alignment of SaHcd1 with homologs. Proteins are indicated by their PDB code: SaHcd1, 7sub; FabG from *Acholeplasma laidlawii*, 4nbt (Javidpour *et al.*, 2014); β -keto acyl carrier protein reductase from *Brassica napus* (with NADP⁺), 1edo (Fisher *et al.*, 2000); FabG from *Synechocystis* sp. PCC 6803, 4rzh (Liu *et al.*, 2015); ORF1438, a putative glucose/ribitol dehydrogenase from *Clostridium thermocellum*, 2hq1 (Southeast Collaboratory for Structural Genomics, unpublished work); FabG from *Staphylococcus aureus* in complex with NADPH, 3sj7 (Dutta *et al.*, 2012); OAR from *Bacillus anthracis* (BA3989), 2uvd (Zaccai *et al.*, 2008); FabG from *Listeria monocytogenes* (with NADP⁺), 4jro (Center for Structural Genomics of Infectious Diseases, unpublished work). Secondary structures of SaHcd1 are visually depicted as arrows (β -strands) and helices (α -helices and $3_{10}/\eta$ -helices). $3_{10}/\eta$ -helices are normally considered as an extension of α -helices. Red bars indicate the unresolved loops in the SaHcd1 structure. Sequence alignment was performed by Blossum62 global alignment with free end gaps using *Geneious Prime* 2023.0.4 (<https://www.geneious.com>; Henikoff & Henikoff, 1992). Shading is based on the Blossum62 matrix with a threshold of 1, so residues that are 100% similar are shaded black, those that are 80–100% similar are shaded charcoal and those that are 60–80% similar are shaded gray. The β - α - β region and the corresponding arginine and tyrosine are denoted by brown rectangles and squares, respectively. The glycine-rich motif is represented by a gray rectangle. Residues that were represented by more than one rotamer in the SaHcd1 structure are indicated by blue boxes. The residues of the catalytic tetrad/triad are denoted by pink squares. Notable residues that are not conserved in SaHcd1 are denoted by gold squares and a region of interest is noted by a gold rectangle. The highly conserved NNAG sequence is denoted by a purple rectangle.

of the TGXXXGXG residues in *SaHcd1* are Thr12 and Gly13, Gly17 and Gly19 (Fig. 2, gray rectangle; Fig. 3a). In the crystal

structure of β -ketoacyl-ACP reductase 1 complexed with NADPH from *S. aureus* subsp. *aureus* NCTC 8325 (PDB entry

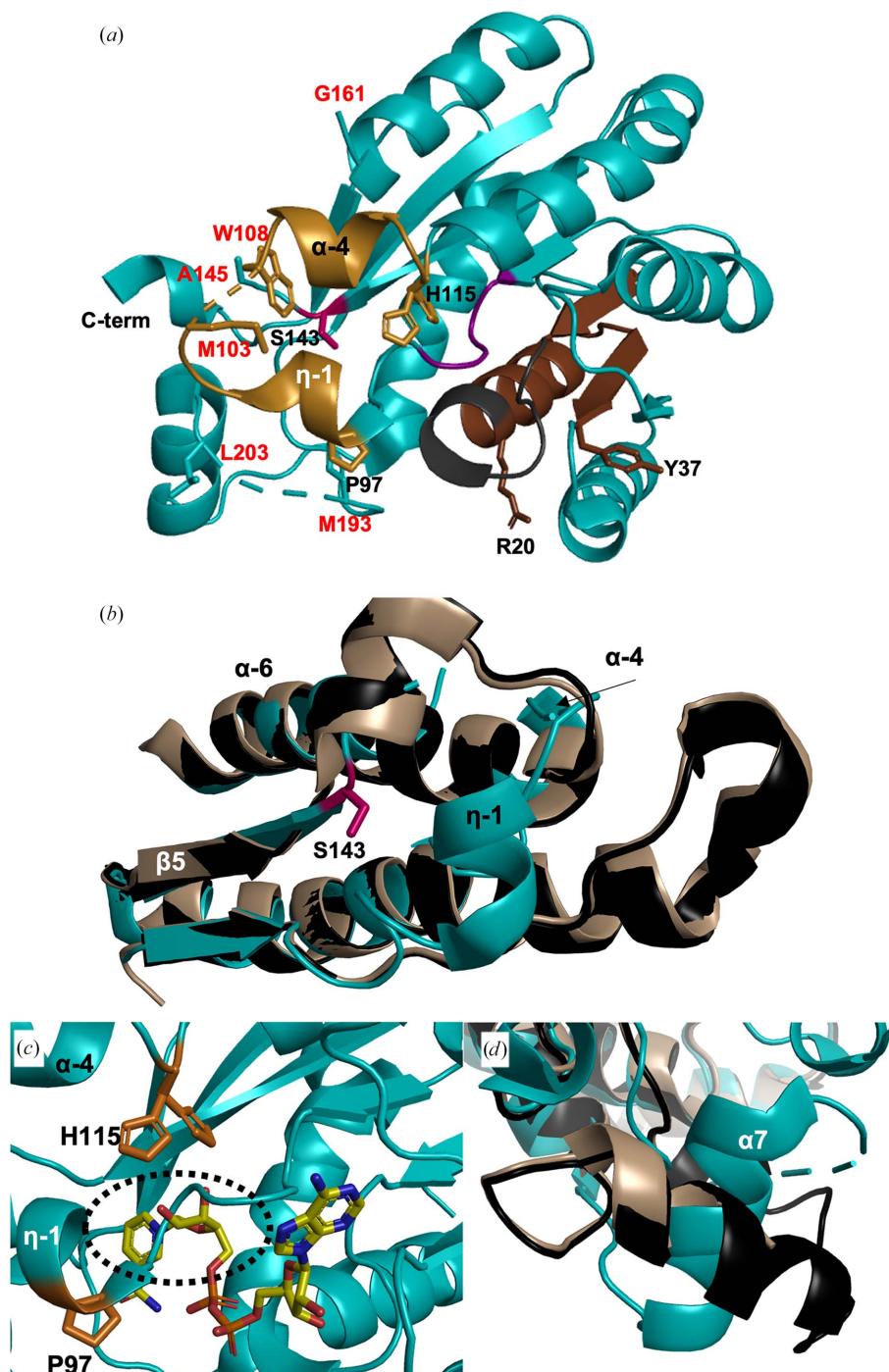


Figure 3

Highlights of the *SaHcd1* structure; significant differences between *SaHcd1* (PDB entry 7sub, cyan) and SDR-family proteins are highlighted. (a) The *SaHcd1* structure is shown in cyan, and features highlighted in the alignment in Fig. 2 are shown along with several residues (Trp108, Ala145, Gly161, Met193 and Leu203 with red labels) flanking the unresolved loops. The β - α - β region is highlighted in brown, and the glycine motif is dark gray/black. The side chains of Arg20 and Tyr37 in this region are shown. The conserved NNAG region is highlighted in purple. The Pro97 and His115 and intervening residues are highlighted in orange. The η -1 and α 4 helices are noted. The Ser143 of the Ser-Tyr-Lys catalytic triad is highlighted in hot pink. OAR from *Bacillus anthracis* (BA3989; PDB entry 2uvd, black; Zaccai *et al.*, 2008) and FabG from *Synechocystis* sp. PCC 6803 (PDB entry 4rzh, gold; Liu *et al.*, 2015) are used as reference structures in (b), (c) and (d). (b) The 3_{10} - η -helix 1 unique to *SaHcd1*, the transition to α 4 and the contrast in this region in *SaHcd1* compared with the comparison SDR-family protein structures are shown. The elongated β 5 strand in *SaHcd1* is visible and the side chain of Ser143 at its C-terminus is shown in hot pink. (c) The side chains of Pro97 and His117 (orange) and their positions relative to η 1 and α 4. The NADP from PDB entry 4nbt is superposed to demonstrate the shift of this region into the cofactor-binding pocket in *SaHcd1* (black dotted circle). His115 has two rotamers in the *SaHcd1* structure. (d) *SaHcd1* α 7 in comparison to the corresponding helix in PDB entries 2uvd and 4rzh.

3sj7), the TGXXXGXXG motif interacts with the adenosine ribose and phosphate moieties of NADPH (Dutta *et al.*, 2012). The *SaHcd1* glycine motif contains a basic residue (Arg16; TGXXRGXXG) and has a tyrosine (Tyr37) positioned at the end of $\beta 2$ (Fig. 2, brown squares; Fig. 3a), which is common among NADPH-binding SDR proteins (Kallberg *et al.*, 2002). NADH-binding SDRs have an aspartate residue at the C-terminal end of the second β -strand of the β - α - β secondary structure at the N-terminus (Fig. 2, brown rectangle) that encompasses the glycine-rich motif (Fig. 2, gray rectangle) (Filling *et al.*, 2002). The negative charge of the aspartate repulses the phosphate group of NADPH, inhibiting binding (Javidpour *et al.*, 2014). *SaHcd1* was initially proposed to concomitantly reduce NAD^+ to NADH while converting 3-hydroxypimelyl-CoA to 3-oxopimelyl-CoA (James *et al.*, 2019). The side chain of Tyr37 is positioned away from the NADPH/NADH-binding pocket of the protein (Fig. 3a), as observed in the *Brassica napus* β -keto acyl carrier protein reductase (PDB entry 1edo) structure with NADP^+ bound. Despite the conservation of the tyrosine residue seen in other NADPH-binding SDR proteins, there is still a possibility that *SaHcd1* can bind to NAD^+ /NADH rather than NADP^+ /NADPH because there is no indication that NAD^+ /NADH binding is structurally unfavorable. Nonetheless, future enzymatic assays of *SaHcd1* should test both NAD^+ /NADH and NADP^+ /NADPH.

A large number of SDR-family proteins contain an extended catalytic tetrad Asn–Ser–Tyr–Lys (Filling *et al.*, 2002). However, *SaHcd1* only contains the central Ser–Tyr–Lys catalytic triad composed of residues Ser143, Tyr156 and Lys160 (Fig. 2, pink squares). The asparagine, while well conserved in closely related SDR proteins used for comparison in this study, is replaced by histidine in *SaHcd1* (Fig. 2, gold squares; Figs. 3a and 3c). The region housing the catalytic triad Ser–Tyr–Lys in *SaHcd1* was not fully resolved (Fig. 2, red bars), with only the serine residue (Ser143) of the triad present in the model. Ser143 sits at the C-terminal tip of the $\beta 5$ strand (Fig. 3a and 3b), which is elongated by three amino acids compared with the SDR-family proteins used in this study (Figs. 2 and 3b, Supplementary Fig. S3), marking the third distinct difference.

It has been suggested that the multimeric state of FabG proteins may dictate their responsiveness to NADPH (Dutta *et al.*, 2012). FabG tetramers have been shown to have negative cooperativity for NADPH (Karmodiya & Suroliya, 2006; Price *et al.*, 2004), whereas *Mycobacterium tuberculosis* FabG1 exists as a dimer and exhibits positive homotropic cooperativity for NADPH (Silva *et al.*, 2008; Marrakchi *et al.*, 2002). Several SDR proteins used for comparison in this study were crystallized in a multimeric state (Supplementary Table S1). SEC (Supplementary Fig. S1a), and PISA analysis (*Protein Interfaces, Surfaces and Assemblies*, Protein Data Bank in Europe; Krissinel & Henrick, 2007) indicated that the biological unit of *SaHcd1* is a dimer; this dimer can also be seen in the crystallographic data through symmetry expansion (Supplementary Fig. S2). As indicated by PISA, the *SaHcd1* dimer interface surface area is $1075.7 \text{ \AA}^2$ and is stabilized by

ten hydrogen bonds and two salt bridges that involve seven amino-acid residues in the C-terminus (Fig. 3, Supplementary Table S2). The complex-formation significance score (CSS) for the interface was only 0.299, which may explain the presence of multiple peaks in SEC, further supporting a monomer–dimer equilibrium. The other SDR-family proteins used for comparison had CSS values from 0.524 to 1.0 and interface areas ranging from $1213.5 \text{ \AA}^2$ for PDB entry 2hq1 to $1652 \text{ \AA}^2$ for PDB entry 4nbt. The interfaces of PDB entry 2hq1 and PDB entry 1edo are stabilized by 14 and 12 hydrogen bonds, respectively, whereas the remaining comparison structures were like *SaHcd1* and were stabilized by both hydrogen bonds and salt bridges. However, unlike *SaHcd1*, the interfaces of other SDR-family proteins were stabilized by a larger number of hydrogen bonds (13–26) and salt bridges (6–8).

The dimerization interface for *SaHcd1* is not shared with all of the SDR-family proteins used for comparison. The *SaHcd1* dimerization interface is located between residues 227 and 253 in the far C-terminal region of the protein (Supplementary Table S2 and Supplementary Fig. S3). The FabG from *Synechocystis* sp. PCC 6803 (PDB entry 4rzh) and a putative glucose/ribitol dehydrogenase from *Clostridium thermocellum* (PDB entry 2hq1) share a similar dimerization interface with *SaHcd1* (Supplementary Fig. S3). As previously mentioned, the dimerization interface of PDB entry 2hq1 is stabilized solely by hydrogen bonds. The dimerization interfaces of *SaHcd1*, PDB entry 2hq1 and PDB entry 4rzh share three common hydrogen bonds. They involve the tyrosine and glutamine in the well conserved Y(I/V)TGQ motif and the aspartic acid in the (V/I)(D/N)GG motif at the C-terminus (Fig. 2; residues 239–250). While both *SaHcd1* (PDB entry 7sub) and PDB entry 4rzh are stabilized by salt bridges, they do not involve shared residues. The remaining SDR-family comparison structures utilize a different dimerization interface across the central portion of the protein that includes the region encompassing the catalytic triad/tetrad (Supplementary Fig. S3) as well as the region shown in Fig. 3(a) (amino acids 90–120) that is an area of difference between *SaHcd1* and the other SDR-family proteins. It is worth noting that while this dimerization interface is not utilized by *SaHcd1*, many of the residues involved in hydrogen bonds and salt bridges at the dimerization interface are often conserved in *SaHcd1*, PDB entry 4rzh and PDB entry 2hq1. Therefore, it is not a lack of residue conservation that results in a shift of the dimerization interface. Still, a further understanding of the physiological roles of each family member is necessary to glean whether there is any correlation between the substrate range of the enzyme and the dimerization interface.

4. Conclusion

Structural insight into *SaHcd1* provides additional insight into SDR-family proteins. Overall, *SaHcd1* bears unique sequence and structural features compared with other SDR-family proteins. Co-crystallization of *SaHcd1* with a cofactor or substrate is an important next step to stabilize the disordered

regions in the hope of obtaining more concrete structural information on SaHcd1.

Acknowledgements

Special thanks to Dr Philip Bourne from the OU PPCC for assistance with size-exclusion chromatography. Thanks to Dr Michael McInerney and Neil Wofford at the University of Oklahoma for *S. aciditrophicus* DNA and consultation on syntrophic metabolism. Thanks to Eric Moore for early contributions to the project.

Funding information

The SaHcd1 project is supported by DOE Basic Energy Sciences Physical Biosciences Program Award DE-FG02-96ER20214 to Dr Elizabeth A. Karr, Associate Professor, and Dr Michael McInerney, Professor Emeritus, University of Oklahoma. The OU MCL was used for crystallization and data collection. NIH COBRE Grant P20GM103640 funds both the OU PPCC and MCL. Use of the SSRL SLAC National Accelerator Laboratory is supported by the DOE Basic Energy Sciences under Contract No. DE-AC02-76SF00515. The SSRL Structural Molecular Biology Program is supported by the DOE Biological and Environmental Research and by the NIH NIGMS (including P41GM103393).

References

- Bhatnagar, B. S., Bogner, R. H. & Pikal, M. J. (2007). *Pharm. Dev. Technol.* **12**, 505–523.
- DeLano, W. L. (2002). *CCP4 Newsl. Protein Crystallogr.* **40**, 82–92.
- Dutta, D., Bhattacharyya, S. & Das, A. K. (2012). *Proteins*, **80**, 1250–1257.
- Elshahed, M., Bhupathiraju, V., Wofford, N., Nanny, M. & McInerney, M. (2001). *Appl. Environ. Microbiol.* **67**, 1728–1738.
- Elshahed, M. S. & McInerney, M. J. (2001). *Appl. Environ. Microbiol.* **67**, 5520–5525.
- Emsley, P. & Cowtan, K. (2004). *Acta Cryst.* **D60**, 2126–2132.
- Emsley, P., Lohkamp, B., Scott, W. G. & Cowtan, K. (2010). *Acta Cryst.* **D66**, 486–501.
- Evans, W. C. (1977). *Nature*, **270**, 17–22.
- Filling, C., Berndt, K. D., Benach, J., Knapp, S., Prozorovski, T., Nordling, E., Ladenstein, R., Jörnval, H. & Oppermann, U. (2002). *J. Biol. Chem.* **277**, 25677–25684.
- Fisher, M., Kroon, J. T. M., Martindale, W., Stuitje, A. R., Slabas, A. R. & Rafferty, J. B. (2000). *Structure*, **8**, 339–347.
- Gabrielli, F., Antinucci, M. & Tofanelli, S. (2023). *Genes*, **14**, 110.
- Henikoff, S. & Henikoff, J. G. (1992). *Proc. Natl Acad. Sci. USA*, **89**, 10915–10919.
- Holm, L. (2020). *Protein Sci.* **29**, 128–140.
- James, K. L., Kung, J. W., Crable, B. R., Mouttaki, H., Sieber, J. R., Nguyen, H. H., Yang, Y., Xie, Y., Erde, J., Wofford, N. Q., Karr, E. A., Loo, J. A., Ogorzalek Loo, R. R., Gunsalus, R. P. & McInerney, M. J. (2019). *Environ. Microbiol.* **21**, 1833–1846.
- Javidpour, P., Pereira, J. H., Goh, E.-B., McAndrew, R. P., Ma, S. M., Friedland, G. D., Keasling, J. D., Chhabra, S. R., Adams, P. D. & Beller, H. R. (2014). *Appl. Environ. Microbiol.* **80**, 497–505.
- Kabsch, W. (2010). *Acta Cryst.* **D66**, 125–132.
- Kallberg, Y., Oppermann, U., Jörnval, H. & Persson, B. (2002). *Eur. J. Biochem.* **269**, 4409–4417.
- Kallberg, Y., Oppermann, U. & Persson, B. (2010). *FEBS J.* **277**, 2375–2386.
- Karmodiya, K. & Suroli, N. (2006). *FEBS J.* **273**, 4093–4103.
- Kavanagh, K. L., Jörnval, H., Persson, B. & Oppermann, U. (2008). *Cell. Mol. Life Sci.* **65**, 3895.
- Krissinel, E. & Henrick, K. (2007). *J. Mol. Biol.* **372**, 774–797.
- Liebschner, D., Afonine, P. V., Baker, M. L., Bunkóczi, G., Chen, V. B., Croll, T. I., Hintze, B., Hung, L.-W., Jain, S., McCoy, A. J., Moriarty, N. W., Oeffner, R. D., Poon, B. K., Prisant, M. G., Read, R. J., Richardson, J. S., Richardson, D. C., Sammito, M. D., Sobolev, O. V., Stockwell, D. H., Terwilliger, T. C., Urzhumtsev, A. G., Videau, L. L., Williams, C. J. & Adams, P. D. (2019). *Acta Cryst.* **D75**, 861–877.
- Liu, Y., Feng, Y., Cao, X., Li, X. & Xue, S. (2015). *FEBS Lett.* **589**, 3052–3057.
- Marrakchi, H., Ducasse, S., Labesse, G., Montrozier, H., Margeat, E., Emorine, L., Charpentier, X., Daffé, M. & Quémard, A. K. (2002). *Microbiology*, **148**, 951–960.
- Mouttaki, H., Nanny, M. A. & McInerney, M. J. (2007). *Appl. Environ. Microbiol.* **73**, 930–938.
- Mouttaki, H., Nanny, M. A. & McInerney, M. J. (2008). *Environ. Microbiol.* **10**, 3265–3274.
- Persson, B., Kallberg, Y., Bray, J. E., Bruford, E., Dellaporta, S. L., Favia, A. D., Duarte, R. G., Jörnval, H., Kavanagh, K. L., Kedishvili, N., Kisiela, M., Maser, E., Mindnich, R., Orchard, S., Penning, T. M., Thornton, J. M., Adamski, J. & Oppermann, U. (2009). *Chem. Biol. Interact.* **178**, 94–98.
- Persson, B., Krook, M. & Jörnval, H. (1991). *Eur. J. Biochem.* **200**, 537–543.
- Price, A. C., Zhang, Y.-M., Rock, C. O. & White, S. W. (2001). *Biochemistry*, **40**, 12772–12781.
- Price, A. C., Zhang, Y.-M., Rock, C. O. & White, S. W. (2004). *Structure*, **12**, 417–428.
- Sambrook, J. (2001). *Molecular Cloning: A Laboratory Manual*, 3rd ed. New York: Cold Spring Harbor Laboratory Press.
- Silva, R. G., Rosado, L. A., Santos, D. S. & Basso, L. A. (2008). *Arch. Biochem. Biophys.* **471**, 1–10.
- Strambini, G. B. & Gabellieri, E. (1996). *Biophys. J.* **70**, 971–976.
- Zaccari, N. R., Carter, L. G., Berrow, N. S., Sainsbury, S., Nettleship, J. E., Walter, T. S., Harlos, K., Owens, R. J., Wilson, K. S., Stuart, D. I. & Esnouf, R. M. (2008). *Proteins*, **70**, 562–567.